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SYSTEMS FOR REDUCING FRICTION IN ENDOVASCULAR DEVICES

Abstract

An endovascular device for navigation of a blood vessel is disclosed. The endovascular device includes a tube with a selectively bendable portion, a control element extending through at least a portion of the tube, wherein movement of the control element relative to the tube is configured to cause the selectively bendable portion of the tube to bend, and a spacer disposed within the tube and placed between the tube and the control element, the spacer comprising at least one wire formed in a coil and configured to reduce sliding friction between the tube and the control element, and wherein the coil has a pitch at least 1.2 times a diameter of the wire.

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Background/Summary

BACKGROUND

[0001] This disclosure relates to the field of endovascular medical devices. Specifically, this disclosure is related to endovascular devices intended to pass through a blood vessel of a patient to a target area inside the patient's body to perform a medical procedure.

[0002] An example of an endovascular treatment of the type relevant to this disclosure is the use of an endovascular device such as a catheter to treat narrowing, blockage, or hemorrhage in a blood vessel, including neurovascular, cardiovascular, and peripheral vasculatures. For instance, treatment of an acute stroke caused by a blockage of a blood vessel in the brain typically comprises either the intra-arterial administration of thrombolytic drugs such as recombinant tissue plasminogen activator (rtPA), mechanical removal of the blockage, or a combination of the two. These interventional treatments must occur within hours of the onset of symptoms. Both intra-arterial (IA) thrombolytic therapy and interventional thrombectomy involve accessing the blocked cerebral artery via endovascular techniques and devices.

[0003] Devices employed in these endovascular procedures may include balloons, snares, coils, stents, temporary stents (including those referred to as “stent-retrievers”), suctioning (e.g., of a blood clot with or without adjunct disruption of the clot (e.g., by use of aspiration catheters)), or a combination of these, depending on the condition to be treated and its anatomical location. Guide catheters, guide sheaths or guidewires are typically used to introduce, guide and/or position interventional devices within the peripheral, cardiovascular and neurovasculature systems from an arterial access site, typically the femoral or radial arteries. These medical devices are often used in a nested fashion, namely, a guidewire inside a microcatheter inside an intermediate catheter are advanced as an assembly to the target site.

[0004] Notably, for an endovascular device to work effectively, the device needs to be positioned as close as possible to the source of disruption, e.g., to a blood clot or a site of narrowing blood vessels. This may be challenging in anatomical areas in which the blood vessels comprise tortuous anatomy, such as the brain. Existing guidewire devices often require the medical practitioner to manually alter a curvature of a tip of the device before the device is inserted into the blood vessel, require the medical practitioner to replace the device by another more suitable guidewire during the procedure, or to remove the guidewire reshape it and reintroduce into the patient. Furthermore, these devices often have difficulty overcoming distal tortuous anatomy due to the lack of support or stiffness, especially for the distal ends of catheters, and lack of the ability to navigate through these rough vessels.

BRIEF SUMMARY OF THE INVENTION

[0005] In a first embodiment, an endovascular device includes a tube comprising a selectively bendable portion, wherein at least a portion of the tube is configured to be inserted into a blood vessel; a plug disposed at a distal end of the tube, wherein the plug has a rounded tip configured to be inserted into the blood vessel; a washer disposed between the plug and the tube, the washer fixed with respect to the tube; and a control element extending through at least a portion of the tube, a distal end of the control element connected to the washer, wherein movement of the control element relative to the tube is configured to cause the selectively bendable portion of the tube to bend.

[0006] In a second embodiment an endovascular device includes a tube comprising a selectively bendable portion; a plurality of slots formed in the tube, each of the slots disposed partially around the circumference of the tube; a control element extending through the tube, the control element connected to a distal portion of the selectively bendable portion, wherein movement of the control element with respect to the tube is configured to bend the selectively bendable portion; and a spacer

disposed within the tube and placed between the tube and the control element, the spacer configured to reduce sliding friction between the tube and the control element.
[0007] A method for controlling stiffness of an endovascular device according to an embodiment of the present disclosure begins with receiving a medical image captured during an endovascular procedure. The medical image is analyzed to detect a target area where a path of a vascular system to a target vessel changes direction. The medical image is also evaluated to detect an endovascular device extending along the path. It is then determined if a particular portion of the endovascular device is positioned at the target area, and the curvature of the particular portion of the endovascular device is controlled as needed.
[0008] A method of navigating a catheter through a blood vessel according to an embodiment of the present disclosure includes steering an endovascular device configured as a guidewire through the blood vessel to a location of interest, wherein the endovascular device comprises a selectively bendable portion at a distal end of the endovascular device and positioning the endovascular device at the location of interest, wherein the location of interest comprises a position along a blood vessel. The method further includes orienting the distal end of the endovascular device by actuating the selectively bendable portion of the endovascular device to create a tip shape; and navigating the catheter through the blood vessel by advancing the catheter over the guidewire to the location of interest.
[0009] Certain aspects of the disclosure have other steps or elements in addition to or in place of those mentioned above. The steps or elements will become apparent to those skilled in the art from a reading of the following detailed description when taken with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS/FIGURES
[0010] The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles thereof and to enable a person skilled in the pertinent art to make and use the same.
[0011] FIG. **1** is a perspective view of an endovascular device according to an embodiment.
[0012] FIG. **1**A is a perspective view of an endovascular device according to an embodiment.
[0013] FIG. **2** is a perspective view of a distal portion of an endovascular device according to an embodiment.
[0014] FIG. **2**A is a side view of a portion of an endovascular device according to an embodiment.
[0015] FIG. **2**B is a side view of a portion of an endovascular device according to an embodiment.
[0016] FIG. **3** is a partial section view of the endovascular device of FIG. **2**.
[0017] FIG. **3**A is a top and side view of an actuator of an endovascular device according to an embodiment.
[0018] FIG. **3**B is a partial section view of the endovascular device according to an embodiment.
[0019] FIG. **3**C is a side view of an exemplary spacer for an endovascular device according to an embodiment.
[0020] FIG. **4** is a cross section of the endovascular device of FIG. **2**.
[0021] FIG. **4**A is a cross section schematic of an endovascular device according to an embodiment.
[0022] FIG. **4**B is a cross section view of the endovascular device of FIG. **3**B.
[0023] FIG. **4**C is a top and side view of an actuator of an endovascular device according to an embodiment.
[0024] FIG. **5** is a perspective view of a proximal portion an endovascular device according to an

embodiment.
[0025] FIG. **6** is cross section of the endovascular device of FIG. **5**.
[0026] FIG. **7** is a partial section view of a proximal portion of an endovascular device according to an embodiment.
[0027] FIG. **7**A is a detail view of a portion of an endovascular device according to an embodiment.
[0028] FIG. **8** is a perspective view of an endovascular device according to an embodiment.
[0029] FIG. **9** is a partial section view of the endovascular device of FIG. **8**.
[0030] FIG. **10** is a cross section of a proximal portion of an endovascular device according to an embodiment.
[0031] FIG. **11** is a diagram of a flowchart showing a method of using an endovascular device according to an embodiment.
[0032] FIG. **12** is a diagram of a flowchart showing a method of using an endovascular device according to an embodiment.
[0033] In the drawings, like reference numbers generally indicate identical or similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

[0034] Reference will now be made in detail to representative embodiments illustrated in the accompanying drawings. References to “one embodiment,” “an embodiment,” “an exemplary embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such a feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.
[0035] Endovascular devices used to perform intravascular medical treatments need to be able to navigate through a blood vessel to reach the treatment site. This can be difficult in situations where the blood vessels in question comprise multiple turns because the typical endovascular device is not able to actively bend to navigate a twist or turn. Thus, an embodiment of the present disclosure is an endovascular device having a tube with a selectively bendable portion, where at least a portion of the tube is configured to be inserted into a blood vessel, and a control element configured to control a curvature of the selectively bendable portion to enable navigation of the tube through the blood vessel.
[0036] The bendable nature of the endovascular device has several benefits, including an improved ability to navigate curved blood vessels and to selectively and controllably change the shape of a portion of the endovascular device from a proximal location, which can be beneficial during certain endovascular procedures. Additional benefits are discussed below.
[0037] As seen in FIG. **1**, an endovascular device **1** is formed as an approximately tubular device. Endovascular device **1** includes a distal end **2** and a proximal end **3** (also referred to as “distal tip” and “proximal tip”, respectively). Proximal end **3** is the end of tube **100** that generally remains outside of a patient and is manipulated by a user (e.g., a doctor or other medical professional). Distal end **2** is the end of tube **100** that is generally inserted into a blood vessel of the patient. In some embodiments, endovascular device **1** may have an overall length of between 1500 and 2500 millimeters (mm). In some embodiments, endovascular device **1** may have an overall length of about 2000 mm.
[0038] Endovascular device **1** is formed from a tube **100**. Tube **100** makes for a body of endovascular device **1**. As seen in FIGS. **1** and **2**, in some embodiments tube **100** can be formed as a cylinder with a central opening. Tube **100** can be formed from any suitable material known in the

art, such as metals or plastics, as discussed in detail below. As tube **100** is intended to be at least partially inserted into a blood vessel, the material of tube **100** should be biocompatible. Examples of materials for tube **100** (e.g., for the entire body of tube **100**, or at least one of the distal end **2**, the proximal end **3**, or the intermediate tubular section including non-bending portion **118**, therebetween, as well as any combinations thereof) are elastic and/or super-elastic polymers, super-elastic metals or various metals/alloys/oxides such, without being limited to, elastomers, silicon polymeric materials like Polydimethylsiloxane (PDMS), silicon adhesives, silicone rubbers, natural rubbers, thermoplastic elastomers, polyamide, polyimide, poly ethylene (PE), poly propylene (PP), polyether etherketone (PEEK), Acrylonitrile butadiene styrene (ABS), epoxys, polytetrafluoroethylene (PTFE), polyurethane, thermoplastic polyurethanes (TPU), Nylon, Polyether block amide (PeBax), Kevlar, stainless titanium, steel or stainless steel, nickel titanium alloy (Nitinol), nickel-chromium alloy, nickel-chromium-iron alloy, cobalt alloy, tungsten, cobalt, chrome, nickel, aluminum, copper, molybdenum or any combination thereof. According to a specific embodiment, tube **100** may be formed from stainless steel. According to some embodiments, tube **100** is configured as a monolithic, or single part, element. In alternative embodiments, tube **100** is configured to combine tubes of different materials (discussed in further detail hereinbelow). In some embodiments, tube **100** is configured to be elastically deformable along at least portions of its length, as will be explained below.

[0039] In some embodiments, tube **100** may have an outer diameter (OD) of between about 0.10 mm and about 1.10 mm. In some embodiments, tube **100** may have an OD of between about 0.30 mm and about 0.90 mm. In specific embodiments, tube **100** may have an OD of about 0.35 mm, about 0.45 mm, about 0.55 mm, about 0.65 mm, about 0.75 mm or about 0.80 mm. As mentioned above, tube **100** can be formed as a cylinder with a central opening. According to some embodiments, tube **100** may have an inner diameter (ID) of between about 0.10 mm and about 0.90 mm. In some embodiments, tube **100** may have an ID of between about 0.20 mm and about 0.70 mm. In specific embodiments, tube **100** may have an inner diameter of about 0.20 mm, about 0.30 mm, about 0.40 mm, about 0.50 mm or about 0.60 mm.

[0040] Endovascular device **1** includes at least one selectively bendable portion **110** (shown in the bent configuration in FIG. **1**). As shown in FIG. **1**, the bendable portion can be located at distal end **2**, however, the bendable portion can be located at any other location along tube **100**, depending on the needs and circumstances. The bendable portion is controlled by an actuator control **200** at proximal end **3**, as discussed in detail herein below. As will be explained in detail below, embodiments such as those shown in FIG. **1**A can have multiple selectively bendable portions **110**.

[0041] Selectively bendable portion **110** is a portion of tube **100** that can be controllably bent with respect to the original axis of tube **100**. As will be discussed further below, selectively bendable portions **110** can be controlled by a user to improve navigation of endovascular device through the blood vessel of a patient. In some embodiments, selectively bendable portion **110** can extend between about 5 mm and about 50 mm in the axial direction along tube **100**. In some embodiments, selectively bendable portion **110** can extend between about 10 mm and about 30 mm in the axial direction along tube **100**. In specific embodiments, selectively bendable portion **110** can extend about 12 mm, about 13 mm, about 14 mm, about 15 mm, about 16 mm, about 17 mm, about 18 mm, about 19 mm, about 20 mm or about 25 mm in the axial direction along tube **100**.

[0042] Actuator control **200** is a control element that enables a user to achieve control of selectively bendable portion **110**. Actuator control **200** includes a handle **210** and a grip **212** for controlling the curvature of selectively bendable portion **110**.

[0043] Grip **212** is configured to allow a user grasp endovascular device **100**. When the user moves the grip **212** linearly with respect to the patient, the endovascular device **100** can move deeper or shallower in a patient's blood vessel. When the user moves grip **212** torsionally, the tube **100** can rotate within a patient's blood vessel, effectively rotating an orientation of selectively bendable portion **110**.

[0044] A user can cause a selectively bendable portion **110** to curve, for example, by changing a relative distance between handle **210** and grip **212**. An interventional radiologist, for example, may grasp handle **210** with one hand and grip **212** with the other. By increasing the relative distance, selectively bendable portion **110** becomes more curved, e.g., a radius of curvature becomes shorter. And, by decreasing the relative distance, selectively bendable portion **110** becomes less curved, e.g., a radius of curvature becomes longer, until selectively bendable portion **110** is a straight extension of tube **100**. In this way, a user, such as an interventional radiologist, is able to adjust selectively bendable portion **110** according to upcoming conditions in the patient's vascular pathway. How this curvature control may be achieved is described below with respect to FIGS. **5-8**.

[0045] For example, when an artery or vein branches out in two directions, the user may move handle **210** with respect to grip **212** to create the desired curvature. Then the user may rotate grip **212** to orient the curvature of selectively bendable portion **110** towards the desired branch and may move grip **212** linearly to move distal end **2** of endovascular device **1** down the desired branch. In this way, the user can navigate distal end **2** of endovascular device **1** to a desired position within a patient's vascular system.

[0046] In some embodiments, as seen in FIGS. **2-3**, openings **102** are formed in tube **100**. As mentioned above, tube **100** may be made of any suitable material, including medical grade stainless steel. Openings **102** are portions removed from the material (e.g., cutouts). In some embodiments, these openings **102** are formed as slots in tube **100** that penetrate at least partially around a circumference of tube **100** from an exterior of tube **100** to an interior of tube **100**. In some embodiments, openings **102** penetrate from the exterior of tube **100** to the interior of tube **100**. In these embodiments, when endovascular device **1** is inserted into a patient's blood vessel, fluid may penetrate through openings **102**. As shown in FIG. **2**, openings **102** do not extend around the entire circumference of tube **100**. Openings **102** can extend, for example, from between ten percent to ninety percent of the circumference of tube **100**.

[0047] In some embodiments, openings **102** can be formed as a regular, rectangular shape projected onto the surface of tube **100** (as shown in FIG. **2**). In these embodiments, for example, openings **102** can have a width in the axial direction of between about 0.01 mm and about 0.1 mm. According to specific embodiments, openings **102** can have a width in the axial direction of between about 0.02 mm and about 0.06 mm. Openings **102** can also be formed from different shapes projected onto the surface of tube **100**, such as slits, circles, ovals, triangles, or any other desired polygonal shape. Openings **102** may be etched from tube **100**, for example, using any manufacturing methods known in the art, such as by cutting or grinding, for example, with a disc, with a semiconductor dicing blade or using laser cutting.

[0048] By removing material from tube **100**, openings **102** reduce the stiffness or resistance to bending of tube **100**. Thus, openings **102** can be used to create more flexible or bendable portions of tube **100**. These are portions of tube **100** that are more susceptible to being controllably bent, as will be discussed below. Changing the extent of openings **102** can therefore be used to control flexibility of tube **100**. For example, increasing the extent, by increasing size, number, shape, or density of openings **102** will reduce the stiffness of tube **100**, or increase the tendency to bend. Conversely, reducing the extent by reducing the size, number, shape, or density of openings **102** increases the stiffness of tube **100**, or decreases the tendency to bend. Thus, as shown in FIG. **1**, selectively bendable portion **110** of tube **100** has openings **102**, while a second portion of tube **100** has no openings **102**. Additionally, tube **100** may comprise a third portion (e.g., a non-bending portion **118** as further discussed below) comprising openings **102** but of reduced size, number, or density as compared to the selectively bendable portion **110**. The third portion of tube **100** may be located anywhere along the tube, such as between the selectively bendable portion **110** and the second portion of tube **100**, and provides a degree of softness to the tube at any required location depending on the needs and circumstances of the device.

[0049] It should be noted that tube **100** may be able to bend to some degree without any openings

102 present. But, the addition of openings **102** allows for the creation of targeted selectively bendable portions **110**. In some embodiments, as shown in FIG. **2**, openings **102** have identical dimensions, but have variable spacing between them. For example, openings **102** disposed in selectively bendable portion **110** may have a spacing of about 0.03 mm to about 0.2 mm (e.g., pitch of about 0.06 mm to about 0.3 mm). According to a specific embodiment, openings **102** disposed in selectively bendable portion **110** may have a spacing of about 0.08 mm to about 0.06 mm (e.g., pitch of about 0.12 mm to about 0.16 mm). Openings **102** in a separate, more proximal portion of tube **100** (such as in a third portion discussed above) are smaller and may be spaced further apart, and may have a spacing of between about 0.03 mm to about 3 mm (e.g., pitch of about 0.06 mm to about 3 mm).

[0050] Another example of changing the extent of openings **102** is changes in spacing between openings **102**. Larger spacing between openings **102** would increase the stiffness of tube **100**, while smaller spacing between openings **102** decreases the stiffness of tube **100**. This technique can be used to create variable gradients of stiffness to tailor the stiffness of tube **100** as desired. For example, in some embodiments, the spacing or pitch of openings **102** in selectively bendable portion **110** is smaller at the end of selectively bendable portion **110** that is closest to distal end **2**. In these embodiments, the spacing increases towards the end of selectively bendable portion **110** that is closer to proximal end **3**. In this way, selectively bendable portion **110** is biased to be more flexible nearer distal end **2**, which is the end that is inserted into the blood vessel first. The rate of change in spacing can be constant, or can be variable to tailor the stiffness change as needed. In some embodiments, there are no openings **102** for a predetermined distance from proximal end **3** of tube **100**. This can be desirable to increase the stiffness of the proximal portion of tube **100**. This portion of tube **100** may also be the portion of tube **100** intended to remain outside of the patient during an endovascular procedure.

[0051] In some embodiments, the design or material of tube **100** can also be used to adjust flexibility of tube **100**. For example, the cross-section dimensions or shape of tube **100** may be altered to increase or decrease flexibility. Changing the material of tube **100** can also change flexibility. For example, as shown in FIG. **4**A, two or more portions of tube **100** may be made from different materials, which can be used to alter flexibility. A stiffer material will make the corresponding portion of tube **100** less flexible, for example. This can be useful when flexibility changes are desired without changing external dimensions of tube **100**. These different materials can be joined together at a connection **103**. Connection **103** can be, for example, any suitable connection technique including, but not limited to, adhesives, welding, or use of a mechanical connector such as a ring or sleeve (used alone or in combination with other connecting techniques).

[0052] As seen in FIGS. **1-3**, in some embodiments there is a single selectively bendable portion **110** disposed at distal end **2** of tube **100**. This enables a portion of tube **100** at distal end **2** to be more flexible, and thus selectively bendable. Selectively bendable portion **110** may also be disposed at various other points along tube **100** as needed to increase flexibility and control. Accordingly, selectively bendable portion **110** may be disposed at a distal portion of endovascular device **1** which is not the distal end **2**, such as up to about 150 mm from the distal end **2**. As seen in FIGS. **2-3**, in some embodiments this distally located, selectively bendable portion **110** is terminated at distal end **2** with a plug **112**. Plug **112** acts to finish or seal distal end **2**. Plug **112** is generally an atraumatic and non-sharp tip which typically has a rounded, oval, or similar shape to improve the ability of tube **100** to transit inside a blood vessel in an atraumatic manner. A washer **114** is disposed between selectively bendable portion **110** and plug **112**. Plug **112** and washer **114** can be fixedly connected to tube **100** by any suitable method, including but not limited to, welding, soldering, brazing, adhesive, or mechanical connections such as fasteners or crimping. Both plug **112** and washer **114** can be made from any suitable biocompatible material, including metals, metal alloys/oxides, adhesives, silicones, and plastics, or any combination thereof. Materials opaque to X-rays, such as platinum, gold, tungsten, tantalum or the like, may be incorporated into the plug or washer, to act as a fluoroscopic marker to aid in visualization while navigating the device in a blood vessel.

[0053] In some embodiments, selectively bendable portion **110** is the only portion of tube **100** that can be controllably actuated or bent. Thus, there can be one or more non-bending portions **118** that are disposed throughout tube **100**. These non-bending portions **118** can be disposed immediately adjacent to selectively bendable portion **110**, or can be spaced apart from selectively bendable portion **110**. It should be noted that non-bending portions **118** may, and often will, still have some ability to passively flex and bend depending on the stiffness of tube **100** in non-bending portions **118**. Techniques for tailoring this stiffness are discussed below.

[0054] As seen in FIGS. **3-4**, for example, endovascular device **1** can include a control element **120** (also referred to as an actuator) disposed inside tube **100**. Control element **120** can take the form of any suitably stiff element that can freely slide inside tube **100**. Sufficient stiffness is needed to prevent control element **120** from buckling excessively when it is put in compression (during the straightening of selectively bendable portion **110**). However, control element **120** must also be sufficiently flexible such that it does not render tube **100** too inflexible. The stiffness of control element **120** also affects the flexibility of selectively bendable portion **110**, and thus the portion of control element **120** that transitions through selectively bendable portion **110** must be designed to allow for selectively bendable portion **110** to bend as desired. Control element **120** can be made of any suitable biocompatible material, including metals, metal alloys and plastics, or any combination thereof. According to one embodiment, control element **120** can be a solid wire, a multi-filament wire, a string, or a thread. For example, control element **120** can be a solid wire formed from any material known in the art, such as but not limited to, nitinol alloy, stainless steel, and a plastic material, or any combination thereof. For example, as shown in FIGS. **1-4**, control element **120** can be formed as a solid wire that is sized to fit inside tube **100**. Control element **120** can also take the form of a tube, as will be discussed below.

[0055] The function of control element **120** is to transmit force to selectively bend or straighten selectively bendable portion **110**. This is accomplished by fixing the distal end of control element **120** to the distal end of the selectively bendable portion **110** (or to a different attachment point of the selectively bendable portion **110** depending on the needs and circumstances of the device). According to one embodiment, the distal end of control element **120** is fixed to washer **114**, which is in turn fixed to an end of selectively bendable portion **110**. Because selectively bendable portion **110** may be placed anywhere along tube **100**, washer **114** may be located at different locations along tube **100** depending on where selectively bendable portion **110** begins and ends. As seen in FIGS. **3**, **3**B and **4**A, in an embodiment, control element **120** passes through an opening in washer **114** and curves to fit into a groove **115** in washer **114**. For example, as shown in FIGS. **3** and **3**B, control element **120** extends through washer **114** traveling in an axial direction, and then curves approximately **180** degrees back to fit into groove **115**, forming a u-shaped curve. In this way, control element **120** is fixed both axially and rotationally to washer **114** at distal end **2** of endovascular device **1**. Examples of these types of curves are shown in both FIGS. **3**, **3**B and **4**A. In addition to passing through washer **114** as described, control element **120** may also be secured by adhesives and mechanical techniques such as a friction fit or the formation of additional bends/angles in control element **120**. This manner of connecting control element **120** to washer **114** also improves the connection between tube **100** and control element **120** because washer **114** substantially reduces the possibility of control element **120** separating from its fixed position with respect to tube **100**.

[0056] Control element **120** can otherwise slide freely inside tube **100** to transmit force from actuator control **200** to washer **114** at distal end **2**. From the straightened position shown in FIG. **2**, retraction of control element **120** with respect to tube **100** towards proximal end **3** will result in selectively bendable portion **110** bending to accommodate the shortened length of control element **120** (as seen in FIG. **1**). Conversely, extension of control element **120** towards distal end **2** will result in straightening of selectively bendable portion **110**. More particularly, in an embodiment, movement of handle **210** in a proximate direction may increase tension on control element **120** causing curvature of selectively bendable portion **110**, and movement of handle **210** in a distal direction may reduce tension on control element **120** causing straightening of selectively bendable portion **110**.

[0057] In some embodiments, control element **120** can be configured to increase the tendency for selectively bendable portion **110** to bend in a desired direction or directions. This is accomplished by altering the shape of control element **120** to increase its tendency to bend in a given direction or directions. Thus, control element **120** can have any lateral cross-section (perpendicular to the longitudinal axis) such as round, elliptic, square, rectangular, and the like.

[0058] For example, in embodiments like the one of FIG. **4** control element **120** is formed from a solid wire with a varying cross section. Closer to proximal end **3**, this embodiment of control element **120** has a circular cross section. As control element **120** continues in the distal direction, control element **120** flattens and has a rectangular cross section, as seen in FIG. **4**. The rectangular cross section tends to bend in the direction perpendicular to the long sides of the rectangle, creating a preferred or biased bending direction for selectively bendable portion **110** that control element **120** is disposed within.

[0059] Additionally or alternatively, control element **120** can maintain the same cross section, but change in size to alter bending tendencies. For example, control element **120** may maintain a circular cross section, but may increase or decrease in diameter to increase or reduce stiffness, respectively. This type of control element **120** would not have a biased bending direction because the cross-sectional shape is maintained but would instead have lesser or greater bending tendencies depending on the dimensions of control element **120**. The dimensions of these changed areas of control element **120** can be varied to achieve the desired resistance to bending when combined with the properties of tube **100**. For example, an embodiment of this type of control element **120** can have different first, second, and third diameters at different locations along the length of control element **120**.

[0060] For example, an embodiment of control element **120** is shown in FIG. **3**A, which shows a side view and top view of control element **120**. In this embodiment, control element **120** has a constant diameter portion **120***a*, a tapered portion **120***b*, and a flattened portion **120***c*. Constant diameter portion **120***a* has a cross-section of a predetermined diameter, which may correspond to parts of control element **120** that are closer to proximal end **3** of tube **100** when assembled into endovascular device **1**. Tapered portion **120***b* has a diameter that constantly decreases as distance from constant diameter portion **120***a* increases. Flattened portion **120***c* has a rectangular cross-section (thus, it is flattened when compared to other parts of control element **120**) that increases in width when viewed from the top as distance from tapered portion **120***b* increases. Flattened portion **120***c* can correspond to the selectively bendable portion **110**, and thus can correspond to distal end **2** of tube **100**. A skilled artisan will understand that flexibility of control element **120** will increase in all bending directions in tapered portion **120***b* as distance from constant diameter portion **120***a* increases because of the reduction in diameter. Bending in flattened portion **120***c* is biased towards bending in the direction in and out of the drawing with respect to the top view because of the flattened cross section shape. This embodiment of control element **120** is shown in tube **100** in FIG. **3**B, and a cross-section of FIG. **3**B is shown in FIG. **4**B.

[0061] In some embodiments, constant diameter portion **120***a* ranges between about 0.1 and about 0.45, e.g., about 0.140 and about 0.180 mm in diameter. At its most proximal end, tapered portion **120***b* has the same diameter as constant diameter portion **120***a*. Tapered portion **120***b* can taper down to a diameter of between about 0.065 mm and about 0.2 mm. The taper may be evenly distributed (a linear taper) along the axial length of tapered portion **120***b*, or can be unevenly distributed. Flattened portion **120***c* may have a width dimension that begins at the smallest diameter of tapered portion **120***b* and increases up to the diameter of constant diameter portion **120***a*. As seen in FIG. **3**A, there can be four steps of increasing width, with the first (most proximal) step ranging from about 0.100 mm to about 0.170 mm, the second step ranging from about 0.110 mm to about 0.180 mm, the third step ranging from about 0.120 mm to about 0.190 mm, and the fourth step ranging from about 0.130 mm to about 0.200 mm. Here, the width dimension is being defined as the dimension in the top to bottom direction of FIG. **3**A with respect to the lower top view. It should be understood that there can be more or less steps in flattened portion **120***c* as needed or desired to adjust flexibility. Flattened portion **120***c* can also be flatted and expand in width in a gradual, non-stepwise fashion, as shown in FIG. **4**C.

[0062] Biased bending as discussed above can also be achieved by distributing the open areas caused by openings **102** unevenly with respect to the circumference of tube **100**. For example, making openings **102** larger on one side of tube **100** would bias tube **100** to bend in the direction of that side. Thus, having asymmetric slots having openings with a greater extent (e.g., more open area) facing in a first radial direction and slots with a smaller extent (e.g., less open area) facing in a second radial direction (e.g., direction opposite the first radial direction), biases the bending of the selectively bendable portion **110** in the first direction when the control element **120** is moved by actuator control **200**. Using any combination of these methods to bias the bending of tube **100** has the advantages of providing predictable bending for the user of endovascular device **1**. This is particularly helpful when distal end **2** of tube **100** has been inserted into a blood vessel and is not visible. The orientation of the remainder of tube **100** (present outside of the blood vessel) allows the user to understand the direction that the distal end of tube **100** with such a bias will bend with respect to the blood vessel.

[0063] In some embodiments, there can be two or more selectively bendable portions of tube **100**. These selectively bendable portions of tube **100** may include biased bending region accomplished by having an asymmetrical distribution of openings **102** as discussed above.

[0064] The same tube **100** having asymmetrical distribution of openings **102** at the one or more selectively bendable portions of tube **100** can also have a symmetrical set of openings disposed in a different portion of tube **100**. These symmetrical openings **100** improve flexibility of tube **100** without introducing any directional bias. Finally, the same tube **100** may have a portion without any openings **102**, for example, on the portion of tube **100** intended to remain outside of the patient during the procedure.

[0065] For example, in some embodiments there may be selectively bendable portion **110** disposed at distal end **2** with asymmetrical openings as discussed above. Moving proximally from selectively bendable portion **110**, there is a non-bending portion **118** that has openings **102** disposed symmetrically about the circumference of tube **100** to increase flexibility of tube **100**. In this embodiment, openings **102** in non-bending portion **118** have an increasing spacing or pitch in the axial direction as non-bending portion **118** progresses axially towards proximal end **3**. This has the effect of gradually increasing stiffness of tube **100** in the proximal direction. Finally, a portion of tube **100** that is proximal to non-bending portion **118** has no openings **102**. These features provide the endovascular device with a soft, flexible and controllable distal section which is optimal for navigation in torturous anatomy, while providing support, shape retention and maintaining optimum torque transmission and tailored pushability in the more proximal sections.

[0066] In some embodiments, as shown in FIG. **2** and discussed above, the symmetrical openings **102** are formed as a slot (e.g., rectangular slot) that is projected or wrapped around some portion of the circumference of tube **100**. In these embodiments, this portion may be anywhere between 0 and 350 degrees of the circumference of tube **100**. The slot that is projected can be between 0.01 mm and 0.1 mm in width, and the pitch or spacing between slots (openings **102**) can be between 0.04 mm and 50 mm. The orientation of the openings **102** can also be varied in different ways. As shown in FIG. **2**, the ends of the openings **102** can be located at varying positions in terms of rotation about the axis of tube **100**. This means that the solid portion of tube **100** between the ends of each opening **102** can be offset rotationally with respect to adjacent openings **102**, as shown in

FIGS. **2**A and **2**B. In some embodiments, this results in openings **102** being interleaved rotationally. In some embodiments, this offset can be arranged in a pattern such that each opening **102** is rotated a specific predetermined angular amount with respect to adjacent openings **102**. According to one embodiment, each opening **102** may be rotated 0 to 350 degrees, for example, about 45 degrees, about 60 degrees, about 90 degrees, about 120 degrees, about 150 degrees, about 180 degrees, etc., with respect to the adjacent openings **102**, which means that the openings **102** would form a repeating pattern in terms of their rotational alignment. Any desirable symmetrical or asymmetrical rotational alignment of openings **102** is possible. That is, the predetermined amount of rotational offset can be the same or can vary as desired between adjacent openings **102**. In some embodiments, the plurality of slots comprises at least two slots (e.g., at least 3, 4, 5, 6, 7, 8, 9, 10, or more slots) dispersed along the tube, wherein each of the slots is rotationally interleaved with an adjacent slot.

[0067] In other embodiments, opening **102** can be inclined with respect to the axis of tube **100** at an angle between 0 and 90 degrees. This angle can be measured with slots that have at least one elongated or linear side, such as slots that are formed from projected rectangles. That is, the angle is taken as the angle between the elongated side and the axis of tube **100**. As seen in FIG. **2**A, openings **102** can be formed at a desired angle instead of being perpendicular to the axis of tube **100** (as seen in FIG. **2**, for example). Any desired angle can be used to slant openings **102**, and the same discussion above with respect to staggering the solid portions of tube **100** applies here. In these embodiments, this portion may be anywhere between 0 and 350 degrees of the circumference of tube **100**. The slot that is projected can be between 0.01 mm and 0.1 mm in width, and the pitch or spacing between slots (openings **102**) can be between 0.04 mm and 50 mm. In some embodiments, openings **102** can include both inclined and perpendicular orientations in different sections of tube **100**. For example, selectively bendable portion **110** may comprise perpendicular openings, and the non-bending portion **118** may comprise non-perpendicular openings. Alternatively, the selectively bendable portion **110** may comprise non-perpendicular openings, and the non-bending portion **118** may comprise perpendicular openings. Furthermore, different parts of the non-bending portion may comprise alternating perpendicular and non-perpendicular sections. These determinations can be made by a person of ordinary skill in the art, based on the teachings provided herein.

[0068] Movement of control element **120** within tube **100** can cause friction. To reduce the friction generated between control element **120** and tube **100**, endovascular device **100** may include a spacer **130**.

[0069] As seen in FIGS. **3** and **4**, spacer **130** can be disposed between control element **120** and the inside of tube **100**. Spacer **130** acts as a spacer and guide that confines the movement of control element **120** inside tube **100**. In these embodiments, and as discussed in the following paragraphs, spacer **130** is described as a solid material. However, spacer **130** may also include a liquid lubrication element, or may only be a liquid lubrication. For example, an oil can be added to the interior of tube **100** (e.g., near distal end **2**). This oil can reduce friction between control element **120** and tube **100**. Examples of suitable oils can include, without limitation, a silicone oil.

[0070] Spacer **130** is fixed with respect to the interior of tube **100**. Suitable means for fixing spacer **130** to tube **100** include, but are not limited to, welding, soldering, brazing, using adhesives, and mechanical connections. In some embodiments, a connector **132** fixes spacer **130** to tube **100**. In other embodiments, spacer **130** can be fixed to washer **114** using any method known in the art, such as with an adhesive, which in turn means spacer **130** is fixed to tube **100** because washer **114** is fixed to tube **100**. According to an alternative embodiment, spacer **130** is fixed to tube **100** in a location closer to distal end **2**. Accordingly, spacer **130** may be fixed at any location along tube **100** so as to enable longitudinal movement of control element **120**. According to one embodiment, spacer **130** can completely surround control element **120** circumferentially. Alternatively, spacer **130** can only partially surround control element **120** circumferentially. In some embodiments, spacer **130** extends the entire length of tube **100**. In other embodiments, spacer **130** extends only a portion of the axial length of tube **100**. In other embodiments, spacer **130** extends the portion of the selectively bendable portion of tube **100**. In some embodiments, spacer **130** extends 1 to 50 percent of the axial length of tube **100**. In other embodiments, spacer **130** extends 5 to 30 percent of the axial length of tube **100**. In other embodiments, spacer **130** extends 10 to 30 percent of the axial length of tube **100**. Furthermore, spacer **130** may be placed at any section along the tube, e.g. at the distal end of the tube, proximal end of the tube, or anywhere in between.

[0071] In some embodiments, spacer **130** can be formed as a continuous tube of material. In other embodiments, spacer **130** can be formed from a strip or wire that is wrapped in a spiral or coil inside tube **100**. In other embodiments, spacer **130** can be formed from several strips of material or wires that are wrapped in a spiral or coil inside tube **100**. Other possible forms of spacer **130** include separate strips of material running the length of tube **100**, or discrete rings of material separated from each other. Spacer **130** may be formed from a biocompatible material that allows control element **120** to slide with respect to spacer **130**. For example, but not limited to, spacer **130** can be made from a metal, a metal alloy/oxide, a silicone, and a plastic material, or any combination thereof. Exemplary materials of spacer **130** include nitinol, platinum, iridium, polytetrafluoroethylene (PTFE), a fluoropolymer, such as polytetrafluoroethylene.

[0072] In some embodiments, as shown for example in FIG. **3**C, a coil-type spacer **130** is formed from at least one wire shaped into a coil. According to one embodiment, a coil-type spacer **130** is formed from a single wire. According to one embodiment, a coil-type spacer **130** is formed from two or more wires, e.g., from 2, 3, 4 or 5 wires. The wire or wires forming the coil typically have a diameter of between 0.020 mm to 0.200 mm. The coil of spacer **130** typically has a diameter (also referred to as “coil diameter”) of between 0.20 mm to 1.20 mm. The coil of spacer **130** has a pitch, which is the linear distance it takes the coil to complete a single rotation about its central axis and which can be measured by finding the linear distance between the same or common angular point on adjacent coils. Thus, the pitch is usually measured as the center wire to center wire. According to a one embodiment, this pitch can range from 1.2 to 20 times the wire diameter, e.g., 1.5 to 10 times the wire diameter, e.g., 1.5 to 5 times the wire diameter. According to a specific embodiment, coil **130** has a pitch at least 1.2 times a diameter of the wire. According to a specific embodiment, coil **130** has a pitch at least 1.5 times a diameter of the wire. According to a specific embodiment, coil **130** has a pitch at least 2 times a diameter of the wire. According to a specific embodiment, coil **130** has a pitch at least 4 times a diameter of the wire. These measurements, and particularly the pitch, have the benefit of ensuring that spacer **130** allows for bending of tube **100**, particularly in selectively bendable portions **110**. It should also be understood that coil-type spacers **130** may maintain the same dimensions (e.g., wire diameter, coil diameter, and pitch) throughout, or may vary these dimensions to achieve different effects, such as increased or decreased resistance to bending.

[0073] In some embodiments, the material selected for spacer **130** is radiopaque, which means spacer **130** is visible on an x-ray scan, or similar types of medical imaging, when placed in the body. This can be achieved by material selection, or by the addition of an additive or coating to spacer **130**. For example, materials such as gold, platinum, tungsten, tantalum or the like, may be incorporated into spacer **130**, to act as a fluoroscopic marker to aid in visualization. According to one embodiment, spacer **130** is at least partially formed from a material selected from the group consisting of a metal alloy and a fluoropolymer material to ensure spacer **130** is sufficiently radiopaque. In other embodiments, tube **100** can be radiopaque, either by material selection (as discussed above) or by the addition of an additive or coating (discussed below).

[0074] As seen in FIGS. **5-7**, an actuator control **200** is disposed at proximal end **3** of tube **100**. As described above, actuator control **200** allows a user to move control element **120** with respect to tube **100**. This portion of tube **100** and actuator control **200** is generally intended to remain outside of the blood vessel and body of a patient.

[0075] In embodiments, actuator control **200** is formed from a tubular handle **210** that is disposed around proximal end **3** of tube **100**. Handle **210** is configured to slide about the outside of tube **100**. This enables handle **210** to move with respect to grip **212** as described above. Handle **210** extends distally along tube **100** to the distal end of tube **100**. According to an embodiment, handle **210** may extend beyond proximal end **3** of tube **100**.

[0076] Control element **120** exits proximal end **3** of tube **100** into the center of handle **210**, and is fixedly connected to the interior of handle **210** by any suitable means, including but not limited to, welding, soldering, brazing, adhesive, or mechanical connections such as fasteners or crimping. Thus, in some embodiments, control element **120** runs through the entire length of the tube and is connected to proximal end **3** of tube **100**. For example, as seen in FIG. **6**, control element **120** can extend to, and be fixed to, a plug **218** that closes the end of handle **210**. In this way, movement of handle **210** with respect to tube **100** moves control element **120**, which in turn bends selectively bendable portion **110**.

[0077] As mentioned above, grip **212** is disposed around the exterior of tube **100** at a location distal to handle **210**. Grip **212** is fixed to the exterior of tube **100** and is configured to act as a resting place for a hand of a user to better control the movement of handle **210**. Both grip **212** and handle **210** can be formed to have a comfortable, non-slip gripping surface suitable for gripping by hand. Handle **210** may be formed from a composite construction with the portion of handle **210** adjacent to tube **100** formed from a material suitable for sliding against tube **100**, such as a hard plastic, while the exterior-facing portion of handle **210** may be formed from a rubber or foam material for improved grip by a user.

[0078] A stopper **216** can be included as part of actuator control **200**. Stopper **216** functions to stop handle **210** from being advanced too far distally, which is along tube **100**. As seen in FIG. **6**, in some embodiments, stopper **216** is a protrusion that is fixed to tube **100**. This protrusion extends inside handle **210** a predetermined distance. Suitable means for fixing stopper **216** to tube **100** include, but are not limited to, welding, soldering, brazing, using adhesives, and mechanical connections. Friction element **140** contacts stopper **216**, preventing handle **210** from advancing too far because friction element **140** is fixed to control element **120**, which is in turn fixed to handle **210** (by way of plug **218**). Other embodiments of stopper **216** can include structures such as protrusions or rings fixed to the interior of handle **210** that are configured to perform the same function.

[0079] As seen in FIG. **5**, one or more markings **214** can be placed on an exterior of tube **100** such that the user can determine the extent of bending via the alignment of handle **210** and markings **214**. Markings **214** can include a neutral marking that corresponds to a straightened position of selectively bendable portion **110**.

[0080] A key structure **150** can be formed in tube **100** to ensure that rotational torqueing of control element **120** is transmitted to tube **100**. For example, a user may twist handle **210** with respect to the axis of tube **100**, which in turn puts a rotational torque on control element **120**. Key structure **150** ensures that this torque is transmitted to tube **100**, which has the effect of rotating tube **100**, depending on the torque applied by the user. For example, FIG. **7**A shows an embodiment of key structure **150** that comprises material placed inside, and fixed to, tube **100**. The opening in key structure **150** may be shaped to correspond to the shape of a narrow portion **151** of control element **120**. The shape of narrow portion **151** (and corresponding key structure **150**) is selected to prevent rotation of control element **120** with respect to tube **100**. For example, control element **120** may be flattened into a rectangular cross-section at narrow portion **151**. It will be appreciated that any suitable shape may be used to prevent rotation of control element **120**. The material of key structure **150** can be any suitable material known in the art, including but not limited to, metals, plastics (such as polyether etherketone (PEEK)) and composite materials. Key structure **150** can be fixed to tube **100** using any suitable process, including but not limited to, adhesives, welding, soldering, and brazing.

[0081] As seen in FIG. **6**, a friction element **140** can be disposed inside tube **100**. Friction element **140** functions to increase the sliding friction affecting control element **120**. This has at least two benefits. First, it allows the feel of moving control element **120** to be tailored to a desired value. This is important because a friction level that is too low will result in unintended movement of control element **120**. A friction level that is too high can make control element **120** difficult to use with precision. In some embodiments, the desired friction level is approximately 0.5 to 2.0 Newtons, measured as the force needed to overcome sliding friction and move handle **210**. The second advantage of tailoring sliding friction is that it enables control element **120** to be designed to remain in a given position once set. This is important because unwanted movement of control element **120** caused, for example, by movement of endovascular device **1** through a blood vessel can lead to issues such as movement of endovascular device **1** from a given position, such as when a catheter (e.g., an aspiration catheter) is being moved over a guidewire to reach a location of interest.
[0082] As seen in FIG. **6**, in some embodiments, friction element **140** is formed from a tubular sleeve **142** that is placed around control element **120** inside tube **100**. Tubular sleeve **142** is fixed to control element **120**, for example by an adhesive, welding, a friction or press fit. Tubular sleeve **142** is sufficiently rigid to be shaped into different shapes. According to one embodiment, as seen in FIG. **6**, tubular sleeve **142** may form several turns such that there are several distinct contact points **144** between tubular sleeve **142** and the inside of spacer **130** (or, in some embodiments, the inside of tube **100**). These contact points **144** produce friction when control element **120** is moved.
[0083] The friction produced can be tailored by altering the total number of contact points **144**, the surface area of contact at each contact point **144**, and the pressure exerted between contact point **144** and the inside of spacer **130**. For example, adding additional bends to tubular sleeve **142** results in more contact points **144**, thereby increasing friction. Increasing the angle of each bend increases the surface area of contact point **144**, also increasing friction.
[0084] Friction element **140** can be placed anywhere in tube **100**, however it can be desirable to place friction element **140** in the portion of tube **100** that is intended to remain outside of the body (near proximal end **3**). This ensures that the rigidity of friction element **140** does not affect how tube **100** bends inside the body.
[0085] As seen in FIG. **7**, other embodiments of friction element **140** can adjust the friction between handle **210** and the outside of tube **100**. In FIG. **7**, a pair of tubes fixed to the inside of handle **210** from friction element **140**. These tubes slide along the outside of tube **100** to set friction to the desired level. Other embodiments of friction element **140** can include structures such as discrete rings or washers that are fixed to control element **120** and contact the inner surface of spacer **130**. Adjustment of friction forces can be accomplished in the same manner as discussed above by increasing the number or size of the contact points between the rings/washers and spacer **130**. Other embodiments of friction element **140** can include structures such as rings, sleeves, or tubes fixed to the interior of spacer **130** that control element **120** passes through. In some embodiments, these structures can be integrated into spacer **130**, as, for example, a narrow section of spacer **130** that applies friction to control element **120**.
[0086] As seen in FIGS. **1**A and **8-10**, in some embodiments tube **100** comprises multiple selectively bendable portions **110**. For example, there can be one, two, three, four, or more selectively bendable portions **110**. The addition of more selectively bendable portions **110** can improve the ability for endovascular device **1** to navigate blood vessels because it adds more controlled directionality to tube **100**. In some embodiments, as shown in FIG. **8**, one selectively bendable portion **110** is disposed at distal end **2**, as discussed above. A second selectively bendable portion **110** is disposed closer to proximal end **3** of tube **100**. In some embodiments, the two selectively bendable portions **110** are adjacent to each other. In other embodiments, the two selectively bendable portions **110** are separated by a non-bending portion **118** of tube **100**. In some embodiments, the separation is between about 5 mm and about 100 mm in an axial direction.

According to some embodiments, the separation is between about 10 mm to about 50 mm in an axial direction. According to specific embodiments, the separation is of about 5 mm, about 10 mm, about 15 mm, about 20 mm, about 25 mm or about 30 mm in an axial direction. The discussion above with respect to biasing the bending of tube **100** applies equally multiple selectively bendable portions **110**. For example, each selectively bendable portion **110** can be biased in a certain radial direction, which may or may not be the same direction as any other selectively bendable portions **110**, using the techniques discussed above.

[0087] Each selectively bendable portion **110** includes its own control element **120** travelling through tube **100** and fixed to the distal end of the selectively bendable portion **110**. Furthermore, each of the selectively bendable portions **110** may have a spacer **130**, friction element **140**, and actuator control **200** disposed at the proximal end **3**, as seen in FIG. **10**. This enables fully independent control of each selectively bendable portion **110**. In some embodiments, there is a single grip **212** shared between the separate actuator controls **200**. In other embodiments, there are two grips **212** each working in conjunction with one of the two actuator controls **200**.

[0088] In the exemplary embodiment shown in FIGS. **1**A and **8-10**, control element **120** corresponding to the distal selectively bendable portion **110** is a wire (e.g., solid wire with a varying cross section as discussed in detail hereinabove). The wire control element **120** is fixed to the more distal selectively bendable portion **110** using the techniques discussed above. Furthermore, a spacer **130** may be placed between the wire control element **120** and the tube **100**, such a spacer may also be fixed to the tube **100** using the techniques discussed above. Control element **120** corresponding to the more proximal selectively bendable portion **110** is a tube (also referred to as a "second tube") that is disposed between the wire control element **120** and tube **100**. A proximal spacer **130** may also be used between the tube control element **120** and tube **100**. This configuration results in a symmetrical distribution of actuators **120**, which reduces the impact to bending of tube **100**. The tube control element **120** is fixed to the more proximal selectively bendable portion **110** using the techniques discussed above. Furthermore, proximal spacer **130** is fixed to the more proximal selectively bendable portion **110** using the techniques discussed above. However, there is not necessarily a washer **114** associated with the more proximal selectively bendable portion **110**.

[0089] In embodiments where each control element **120** is formed as a wire, each control element **120** is routed through the center of tube **100**, and then fixed to its respective selectively bendable portion **110** using the techniques discussed above. As discussed above, in some embodiments, it is desirable to fix control element **120** to the distal-most end of selectively bendable portion **110**, but different attachment points are possible. In these embodiments, each control element **120** can have its own spacer **130**, or a single spacer **130** can be used as discussed above.

[0090] Actuator controls **200** for devices comprising two or more selectively bendable portions **110** function in the same manner as the single selectively bendable portion **110** discussed above. However, as seen in FIG. **10**, handles **210** for each actuator control **200** are staggered along tube **100** such that they can be independent actuated. For example, handle **210** associated with the proximal selectively bendable portion **110** can be placed more distally along tube **100**. A connection between this handle **210** and its control element **120** can be accomplished via a slot or opening in tube **100**. Handle **210** associated with the proximal selectively bendable portion **110** can be arranged as discussed above. Both handles **210** can have markings **214** configured as discussed above, and also can each have stoppers **216** that act as travel stops as discussed above.

[0091] In some embodiments, at least some of tube **100** may be coated with various substances to improve system performance. For example, an exterior surface of tube **100** intended for insertion into a patient may be entirely or partially equipped with an elastic or otherwise compliant, biocompatible coating or sheath to provide a smooth outer surface hydrophobic or hydrophilic, depending on the needs and circumstances. A coating material is selected to minimize sliding friction of the device during insertion and removal into a subject's body, and is substantially

chemically inert in the in vivo vascular environment. According to one embodiment, the exterior surface of tube **100** may have a hydrophilic coating to reduce friction between tube **100** and a blood vessel. Examples of suitable coatings include, but are not limited to, polytetrafluoroethylene (PTFE), tetrafluoroethylene (TFE), urethane, polyurethane, thermoplastic polyurethanes (TPU), silicone Polyether block amide (PeBax), Nylon or polyethylene (PE), other polymers, polyurethane polymers, and elastomers are also suitable for coating. Additionally or alternatively, the coating material may be selected for its hydrophilic properties thus improving gliding in blood and navigability. Typically, this kind of coating is applied at the distal end **2** of tube **100** and extends up to 50 cm from the tip. Suitable coatings can be formed by any method known in the art, such as by dipping, spraying or wrapping and heat curing operations.
[0092] Other coating options include medications, which can be delivered into the body of the patient during operation of endovascular device **1**. Such drugs can be used for treatment of a patient, to assist the efficacy of the procedure, or to avoid complications capable of evolving as a result of the procedure. For example, coatings of this type can include anti-restenotic drugs (including drugs which inhibit coagulation, such as anticoagulants, antithrombotic agents and antiplatelet agents like clopidogrel and heparin), anti-inflammatory agents, immune suppressants, antibiotics, antihyperlipidemic agents, antiproliferative agents and pro-endothelializing agents. An example of a detailed list of these types of coatings can be found in Tables 8 and 12 of Rykowska I. et al., Molecules (2020) 25:4624; doi:10.3390/molecules25204624, which is incorporated herein by reference.
[0093] In some embodiments, control element **120** can be coated to reduce friction. Suitable coatings for control element **120** include polymers and elastomers. For example, polytetrafluoroethylene (PTFE), tetrafluoroethylene (TFE), nylon, may be used to coat control element **120** to reduce friction when control element **120** moves with respect to tube **100**. Suitable coatings can be formed by any method known in the art, as discussed above.
[0094] As diagramed in FIG. **11**, a method of navigating a catheter begins with a step **1102** when distal end **2** of endovascular device **1** is inserted into a blood vessel by a user, such as a doctor or other medical professional performing a medical procedure. In some embodiments this blood vessel may be spatially separated from a location of interest. For example, the blood vessel may be located in the thigh of a patient or the wrist of a patient, and the location of interest may be located in the head, neck, stomach, chest or peripheral organs of the patient. The location of interest may be any location inside the body requiring a medical treatment to be performed with the assistance of endovascular device **1**. For example, the location of interest may be a location comprising a blood clot, a stenosis, a contracted blood vessel (e.g., a vasospasm) or an aneurysm.
[0095] Next, in a step **1104**, endovascular device **1** is advanced or steered through the blood vessel (or blood vessels) until it is positioned at the location of interest. This is accomplished, for example, by the user manipulating actuator control **200** to advance endovascular device **1**. Selectively bendable portion **110** is controlled via actuator control **200** (which remains outside the patient's body) as needed to guide endovascular device **1** through the various curves present in the blood vessel.
[0096] Once distal end **2** has arrived at the location of interest, in a step **1106** selectively bendable portion **110** located at distal end **2** can be actuated to control the curvature of the tip to create a tip shape that orients the tip as needed. Additionally, or alternatively, the selectively bendable portion **110** located at distal end **2** can be actuated to increase support and/or to provide shape retention to orient the distal end of the endovascular device. For example, the tip shape can be oriented such that it is aligned or aimed at the location of interest. In some embodiments this tip shape can be a bent, or non-straightened shape, a J-shape, or a cobra shape. This orienting of the tip shape does not require any additional support to maintain its orientation. For example, no support is needed from the blood vessel, additional guidewires, additional catheters, or additional endovascular devices.
[0097] In some embodiments, endovascular device **1** is configured to function as a guidewire that

can be used to guide a catheter through the blood vessel to the location of interest. Thus, in a step **1108**, the catheter can be advanced through the blood vessels to perform the relevant medical procedure at the location of interest. For example, the catheter can be an aspiration catheter, and the location of interests can be at or adjacent to a blood clot. Upon arriving at the location of interest, the aspiration catheter can be used to remove the blood clot via suction or other suitable treatment. These embodiments of endovascular device **1** can also be used to guide other medical devices, such as additional guide wires, different types of catheters, or any other suitable systems to locations of interest accessible via a blood vessel in the body. For example, the additional device can be a stent retriever, which can be used to retrieve a blood clot from the blood vessel by applying the stent retriever through the microcatheter. According to another example, the catheter can be a microcatheter, and the location of interest may be a location of a clot, of an aneurysm, or of a narrowed blood vessel, such as a vasospasm or stenosis.

[0098] As diagramed in FIG. **12**, a further method for navigating embodiments of endovascular device **1** through a vascular system begins with a step **1202**, with receiving a medical image captured during an endovascular procedure. This medical image can be taken via any suitable medical imaging technique, such as but not limited to, X-ray, CT or MRI. The image can be stored on a suitable computer or other device.

[0099] In a step **1204**, the medical image is analyzed to detect a target area where a path of the vascular system to a target blood vessel changes direction. For example, this may include a curve or twist in a blood vessel that requires additional steps for endovascular device **1** to navigate successfully. This can be accomplished, for example, by visual examination of the image, or via an appropriate algorithm run by a computer. For example, this may be accomplished with a computer vision model trained to recognize the blood vessel.

[0100] In a step **1206**, the medical image is further analyzed to detect endovascular device **1** extending through the target blood vessel. This can be accomplished, for example, by visual examination of the image, or via an appropriate algorithm run by a computer. This step can be facilitated by the radiopacity of one or more elements of endovascular device **1**, as discussed above. For example, this may be accomplished with a computer vision model trained to recognize radiopaque elements.

[0101] Once both the target area and endovascular device **1** are located, in a step **1208**, a determination is made regarding whether a particular portion of the endovascular device is positioned at the target area. This can be accomplished using techniques similar to those discussed with respect to steps **1206** and **1208**.

[0102] If that is the case, then in a step **1210** curvature of the particular portion of endovascular device **1** is controlled to navigate to the target area. This can be accomplished by altering the curvature of selectively bendable portion **110** as discussed above. The particular portion of endovascular device **1** can be or include distal end **2**, and selectively bendable portion **110** can be located at distal end **2**. In these embodiments, distal end **2** does not transition beyond the target area.

[0103] In some embodiments, endovascular device **1** is configured as a guidewire for guiding a suitable medical device, such as an aspiration catheter to the target area. In other embodiments, endovascular device **1** is itself configured as a catheter (e.g., as a guidecatheter), for example.

[0104] After actions such as a medical treatment are completed, a second medical image can be taken to ensure a successful treatment. In addition, or in the alternative, a signal received from a sensor **108** disposed in endovascular device **1** can be used to determine completion of the action. Sensor **108** can be any suitable sensor, and, for example, can function to sense the presence of a clot, the type of clot, or the type of stenosis present in the blood vessel. After the action is completed, the curvature of endovascular device can be controlled to reverse or modify the initial curvature control step. This allows for removal of endovascular device **1**.

[0105] In embodiments of endovascular device **1** with multiple selectively bendable portion **110**, the initial process described above regarding controlling curvature of the particular portion can be done for each selectively bendable portion **110** using suitable imagery.

[0106] These steps can also be performed on a suitable computing device possessing a memory and a processor. In particular the evaluation of the imagery and determination of the relative positions of endovascular device **1** and the target area can be performed by suitable algorithms stored on memory and run by a processor.

[0107] Exemplary embodiments of the invention are further provided below.

Example 1

[0108] An endovascular device comprising a tube comprising a selectively bendable portion, wherein at least a portion of the tube is configured to be inserted into a blood vessel; a plug disposed at a distal end of the tube, wherein the plug has a rounded tip configured to be inserted into the blood vessel; a washer disposed between the plug and the tube, the washer fixed with respect to the tube; a control element extending through at least a portion of the tube, a distal end of the control element connected to the washer, wherein movement of the control element relative to the tube is configured to cause the selectively bendable portion of the tube to bend; and a spacer disposed within the tube and placed between the tube and the control element, the spacer comprising at least one wire formed in a coil and configured to reduce sliding friction between the tube and the control element, and wherein the coil has a pitch at least 1.2 times a diameter of the wire.

Example 2

[0109] An endovascular device, comprising: a tube comprising a selectively bendable portion, wherein at least a portion of the tube is configured to be inserted into a blood vessel; a plurality of slots formed in the tube, each of the slots disposed partially around the circumference of the tube; a control element extending through at least a portion of the tube, the control element connected to a distal portion of the selectively bendable portion, wherein movement of the control element relative to the tube is configured to cause the selectively bendable portion of the tube to bend; and a spacer disposed within the tube and placed between the tube and the control element, the spacer comprising at least one wire formed in a coil and configured to reduce sliding friction between the tube and control element, and wherein the coil has a pitch at least 1.2 times a diameter of the wire.

Example 3

[0110] The endovascular device of example 2, further comprising a plug disposed at a distal end of the tube, wherein the plug has a rounded tip configured to be inserted into a blood vessel.

[0111] Example 4

[0112] The endovascular device of any one of examples 2 or 3, further comprising a washer disposed in the tube, the washer fixed with respect to the tube, wherein at least one of the control element and the spacer are connected to the washer.

Example 5

[0113] The endovascular device of any one of the preceding examples, wherein the washer is fixed rotationally with respect to an axis of the tube such that application of torque to the control element causes the selectively bendable portion of the tube to bend.

Example 6

[0114] The endovascular device of any one of the preceding examples, wherein a distal end of the spacer is connected to the washer.

Example 7

[0115] The endovascular device of any one of the preceding examples, wherein the control element is configured to slide with respect to at least a portion of the tube.

Example 8

[0116] The endovascular device of any one of the preceding examples, wherein the control element extends through the tube from the washer at the distal end of the tube to a proximal end of the tube.

Example 9

[0117] The endovascular device of any one of the preceding examples, wherein the proximal end of the control element is connected to an actuator control.

Example 10

[0118] The endovascular device of any one of the preceding examples, wherein the diameter of the control element tapers from the proximal end of the tube to the distal end of the tube.

Example 11

[0119] The endovascular device of any one of the preceding examples, wherein a portion of the control element closer to the distal end of the tube has a rectangular cross-section, and a portion of the control element closer to a proximal end of the tube has a circular cross-section.

Example 12

[0120] The endovascular device of any one of the preceding examples, wherein the spacer surrounds at least part of the control element circumferentially.

Example 13

[0121] The endovascular device of any one of the preceding examples, wherein the spacer is at least partially formed from a radiopaque material.

Example 14

[0122] The endovascular device of example **1**, wherein the tube comprises a plurality of slots, each of the slots disposed partially around the circumference of the tube.

Example 15

[0123] The endovascular device of any one of the preceding examples, wherein a first slot of the plurality of slots faces in the first direction, and a second slot of the plurality of slots faces in a second direction different from the first direction, such that a first extent of the first slot is greater than a second extent of the second slot. According to an embodiment, the plurality of slots are formed on a portion of the tube comprising the selectively bendable portion

Example 16

[0124] The endovascular device of any one of the preceding examples, wherein the extent difference between the first slot and the second slot biases the selectively bendable portion to bend in the first direction when the wire is moved.

Example 17

[0125] The endovascular device of any one of the preceding examples, wherein the first slot of the plurality of slots has a larger area than the second slot of the plurality of slots.

Example 18

[0126] The endovascular device of any one of the preceding examples, wherein the first slot of the plurality of slots has a different shape than the second slot of the plurality of slots.

Example 19

[0127] The endovascular device of any one of the preceding examples, wherein the second direction is oriented to face an opposite radial direction than the first direction.

Example 20

[0128] The endovascular device of any one of the preceding examples, further comprising a second plurality of slots formed on a portion of the tube separate from the selectively bendable portion, wherein the second plurality of slots are distributed symmetrically about the tube, and wherein the second plurality of slots enhances the flexibility of the tube in the portion.

Example 21

[0129] The endovascular device of any one of the preceding examples, wherein a first slot of the plurality of slots is rotationally offset with respect to an adjacent second slot of the plurality of slots such that ends of the first slot are not aligned with ends of the second slot.

Example 22

[0130] The endovascular device of any one of the preceding examples, wherein the plurality of slots comprises at least two slots dispersed along the tube, and wherein each of the slots is rotationally interleaved with an adjacent slot.

Example 23
[0131] The endovascular device of any one of the preceding examples, wherein each slot of the plurality of slots is rotationally offset from an adjacent slot of the plurality of slots by a predetermined angle.
Example 24
[0132] The endovascular device of any one of the preceding examples, wherein each slot of the plurality of slots are formed with an elongated side, and wherein the elongated side is aligned perpendicular to an axis of the tube.
Example 25
[0133] The endovascular device of any one of the preceding examples, wherein each slot of the plurality of slots are formed with an elongated side, and wherein the elongated side is aligned at a non-perpendicular angle to an axis of the tube.
[0134] It is to be appreciated that the Detailed Description section, and not the Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present invention as contemplated by the inventor(s), and thus, are not intended to limit the present invention and the appended claims in any way. Moreover, the examples described above do not limit the present disclosure to what has been particularly shown and described hereinabove. Rather, the scope of the present disclosure includes both combinations and sub-combinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art.
[0135] The use of the modifiers “approximately” or “about” in this disclosure are intended to indicate that the relevant element is subject to variation by a tolerance range. Unless otherwise defined, the use of these modifiers with respect to a unit of measure means a tolerance of plus or minus ten percent of the unit of measure. The use of these modifiers with respect to a description such as a shape is intended to allow for variations of that shape due to tolerance issues as would be understood to occur in the art in general.
[0136] The foregoing description of the specific embodiments will so fully reveal the general nature of the invention that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present invention. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.
[0137] Various features of the invention which are, for clarity, described in the contexts of separate embodiments may also be provided in combination in a single embodiment. Conversely, various features of the invention which are, for brevity, described in the context of a single embodiment may also be provided separately or in any suitable sub-combination.
[0138] The breadth and scope of the present invention should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. An endovascular device, comprising: a tube comprising a selectively bendable portion, wherein at least a portion of the tube is configured to be inserted into a blood vessel; a plug disposed at a distal end of the tube, wherein the plug has a rounded tip configured to be inserted into the blood

vessel; a washer disposed between the plug and the tube, the washer fixed with respect to the tube; a control element extending through at least a portion of the tube, a distal end of the control element connected to the washer, wherein movement of the control element relative to the tube is configured to cause the selectively bendable portion of the tube to bend; and a spacer disposed within the tube and placed between the tube and the control element, the spacer comprising at least one wire formed in a coil and configured to reduce sliding friction between the tube and the control element, and wherein the coil has a pitch at least 1.2 times a diameter of the wire.
2. The endovascular device of claim 1, wherein the washer is fixed rotationally with respect to an axis of the tube such that application of torque to the control element causes the selectively bendable portion of the tube to bend.
3. The endovascular device of claim 1, wherein a distal end of the spacer is connected to the washer.
4. The endovascular device of claim 1, wherein the control element is configured to slide with respect to at least a portion of the tube.
5. The endovascular device of claim 1, wherein the control element extends through the tube from the washer at the distal end of the tube to a proximal end of the tube.
6. The endovascular device of claim 5, wherein the proximal end of the control element is connected to an actuator control.
7. The endovascular device of claim 5, wherein a diameter of the control element tapers from the proximal end of the tube to the distal end of the tube.
8. The endovascular device of claim 1, wherein a portion of the control element closer to the distal end of the tube has a rectangular cross-section, and a portion of the control element closer to a proximal end of the tube has a circular cross-section.
9. The endovascular device of claim 1, wherein the tube comprises a plurality of slots, each of the slots disposed partially around the circumference of the tube.
10. The endovascular device of claim 9, wherein a first slot of the plurality of slots faces in the first direction, and a second slot of the plurality of slots faces in a second direction different from the first direction, such that a first extent of the first slot is greater than a second extent of the second slot.
11. The endovascular device of claim 9, further comprising a second plurality of slots formed on a portion of the tube separate from the selectively bendable portion, wherein the second plurality of slots are distributed symmetrically about the tube, and wherein the second plurality of slots enhances the flexibility of the tube in the portion.
12. The endovascular device of claim 9, wherein a first slot of the plurality of slots is rotationally offset with respect to an adjacent second slot of the plurality of slots such that ends of the first slot are not aligned with ends of the second slot.
13. The endovascular device of claim 9, wherein each slot of the plurality of slots are formed with an elongated side, and wherein the elongated side is aligned perpendicular to an axis of the tube.
14. The endovascular device of claim 9, wherein each slot of the plurality of slots are formed with an elongated side, and wherein the elongated side is aligned at a non-perpendicular angle to an axis of the tube.
15. An endovascular device, comprising: a tube comprising a selectively bendable portion, wherein at least a portion of the tube is configured to be inserted into a blood vessel; a plurality of slots formed in the tube, each of the slots disposed partially around the circumference of the tube; a control element extending through at least a portion of the tube, the control element connected to a distal portion of the selectively bendable portion, wherein movement of the control element relative to the tube is configured to cause the selectively bendable portion of the tube to bend; and a spacer disposed within the tube and placed between the tube and the control element, the spacer comprising at least one wire formed in a coil and configured to reduce sliding friction between the tube and the control element, and wherein the coil has a pitch at least 1.2 times a diameter of the

wire.

16. The endovascular device of claim 15, wherein the spacer surrounds at least part of the control element circumferentially.

17. The endovascular device of claim 15, wherein the spacer is at least partially formed from a radiopaque material.

18. The endovascular device of claim 15, wherein a first slot of the plurality of slots faces in the first direction, and a second slot of the plurality of slots faces in a second direction different from the first direction, such that a first extent of the first slot is greater than a second extent of the second slot.

19. The endovascular device of claim 18, wherein the extent difference between the first slot and the second slot biases the selectively bendable portion to bend in the first direction when the wire is moved.

20. The endovascular device of claim 18, wherein the first slot of the plurality of slots has a larger area than the second slot of the plurality of slots.

21. The endovascular device of claim 18, wherein the first slot of the plurality of slots has a different shape than the second slot of the plurality of slots.

22. The endovascular device of claim 18, wherein the second direction is oriented to face an opposite radial direction than the first direction.

23. The endovascular device of claim 15, further comprising a second plurality of slots formed on a portion of the tube separate from the selectively bendable portion, wherein the second plurality of slots are distributed symmetrically about the tube, and wherein the second plurality of slots enhances the flexibility of the tube in the portion.

24. The endovascular device of claim 15, wherein a first slot of the plurality of slots is rotationally offset with respect to an adjacent second slot of the plurality of slots such that ends of the first slot are not aligned with ends of the second slot.

25. The endovascular device of claim 24, wherein the plurality of slots comprises at least two slots dispersed along the tube, and wherein each of the slots is rotationally interleaved with an adjacent slot.

26. The endovascular device of claim 15, wherein each slot of the plurality of slots is rotationally offset from an adjacent slot of the plurality of slots by a predetermined angle.

27. The endovascular device of claim 15, wherein each slot of the plurality of slots are formed with an elongated side, and wherein the elongated side is aligned perpendicular to an axis of the tube.

28. The endovascular device of claim 15, wherein each slot of the plurality of slots are formed with an elongated side, and wherein the elongated side is aligned at a non-perpendicular angle to an axis of the tube.

29. The endovascular device of claim 15, further comprising a plug disposed at a distal end of the tube, wherein the plug has a rounded tip configured to be inserted into a blood vessel.

30. The endovascular device of claim 15, further comprising a washer disposed in the tube, the washer fixed with respect to the tube, wherein at least one of the control element and the spacer are connected to the washer.